Corporate Finance

Topic 5

Bernardo Ricca

Nonverifiable income

- The entrepreneur can divert income (always report failure) without sanctions. Interpretation:
  - Especially for small firms, sales might not be recorded by invoices
  - Difficult to judge whether some expenses are necessary or perks
  - Sovereign debt: lender cannot seize the cashflows (nor collateral) of the borrowing country
- Apart from **liquidity default**, when the borrower is not able to repay because the project has not generated enough cash, the lender has to worry about **strategic default**: borrower is able to repay, but chooses not to do so

Nonverifiable income

- Means to deter strategic defaults:
  - Direct bankruptcy costs, such as collateral seizure or penalties
  - When the lender and the borrower interact repeatedly, threatening termination or nonfinancing of future projects

Threat of termination

- Risk-neutral entrepreneur with wealth A has access to a project that costs I at $t = 0$
- The project generates nonverifiable cashflow $\tilde{R}_1 \in \{0, R_1\}$ at $t = 1$. The probability of success is $Pr(\tilde{R} = R_1) = p$
- There is no discounting
- The project is profitable (has positive NPV): $pR_1 > I$

Threat of termination

- Contracts cannot be contingent on the realization of $\tilde{R}_1$, since it is not verifiable
- After obtaining $I - A$ from a lender, the borrower always reports $\hat{R}_1 = 0$
- The project cannot be financed

Threat of termination

- Suppose now that, apart from the $t = 1$ income, the project generates income $\tilde{R}_2 \in \{0, R_2\}$ at $t = 2$ if it is not terminated at $t = 1$
- The probability of success is $Pr(\tilde{R}_2 = R_2) = p$
- There is nothing the lender can do to incentivize truthful reporting at $t = 2$: borrower will always report $\hat{R}_2 = 0$ and pay 0 to the lender
- However, at $t = 1$, the lenders can incentivize repayment by threatening to liquidate the project
- If the project is liquidated at $t = 1$, the entrepreneur receives 0 at $t = 2$ and lenders receive a liquidation value L at $t = 1$

Threat of termination

- Contract: $\{D, \gamma_0, \gamma_1\}$, where
  - $0 \leq D \leq R_1$ is the repayment at $t = 1$
  - $\gamma_0 \in [0, 1]$ is the probability of continuation if D is not paid
  - $\gamma_1 \in [0, 1]$ is the probability of continuation if D is paid
- Assumptions:
  - **(A1)** Liquidation is inefficient: $L < \mathbb{E}(\tilde{R}_2)$
  - **(A2)** $0 \leq L < I - A$
- If the project is successful at $t = 1$ the borrower truthfully reports $\hat{R}_1 = R_1$ and repays D if

$$R_1 - D + \gamma_1 \mathbb{E}(\tilde{R}_2) \geq R_1 + \gamma_0 \mathbb{E}(\tilde{R}_2) \iff (\gamma_1 - \gamma_0)\mathbb{E}(\tilde{R}_2) \geq D$$

Threat of termination

- The optimal contract solves

$$\max_{\{D,\gamma_0,\gamma_1\}} p\Big[R_1 - D + \gamma_1 \mathbb{E}(\tilde{R}_2)\Big] + (1-p)\Big[\gamma_0 \mathbb{E}(\tilde{R}_2)\Big]$$

subject to

$$(IR_{\text{lender}}) \quad p\Big[D + (1-\gamma_1)L\Big] + (1-p)\Big[(1-\gamma_0)L\Big] \geq I - A$$

$$(IC_{\text{borrower}}) \quad (\gamma_1 - \gamma_0)\mathbb{E}(\tilde{R}_2) \geq D$$

$$(LL) \quad 0 \leq D \leq R_1$$

- **Claim 1:** (IR_{lender}) is binding

  Note that, given (A2), we must have $D > 0$ (and, as a result, $\gamma_1 > \gamma_0$). Then it is possible reduce D by a small amount such that all constraints are satisfied and thus increase the borrower's payoff

Threat of termination

- **Claim 2:** $\gamma_1 = 1$ (there is no liquidation when the borrower repays D)

  Suppose that $1 > \gamma_1 > \gamma_0 \geq 0$. Increase γ_1 by $\epsilon \in (0, 1 - \gamma_1]$ and increase D to $D + \epsilon L$ such that (IR_{lender}) is satisfied. The new contract satisfies the (IC_{borrower})

  $$(\gamma_1 + \epsilon - \gamma_0)\mathbb{E}(\tilde{R}_2) = (\gamma_1 - \gamma_0)\mathbb{E}(\tilde{R}_2) + \epsilon\mathbb{E}(\tilde{R}_2) > D + \epsilon\mathbb{E}(\tilde{R}_2) > D + \epsilon L$$

  The first inequality is due to the fact that the original contract is incentive-compatible, and the second one is due to assumption that liquidation is inefficient: $L < \mathbb{E}(\tilde{R}_2)$. The borrower's payoff increase by $-p\epsilon L + p\epsilon\mathbb{E}(\tilde{R}_2) > 0$.

- Liquidation in case of repayment at $t = 1$ undermines incentives and increases inefficiency, as we assumed that $L < \mathbb{E}(\tilde{R}_2)$

Threat of termination

- **Claim 3:** (IC_{borrower}) is binding

  Note that, since $D > 0$, we must have $\gamma_0 < 1$. Then it is possible to increase γ_0 to $\gamma_0 + \epsilon$ and increase D by $\epsilon L(1-p)/p$ such that (IR_{lender}) is satisfied. By choosing ϵ small enough, (IC_{borrower}) is also satisfied.[1] The borrower's payoff increase by $(1-p)(\mathbb{E}(\tilde{R}_2) - L)\epsilon > 0$

- Given these results, the optimal D and γ_0 solve

$$pD + (1-p)\Big[(1-\gamma_0)L\Big] = I - A$$

$$(1-\gamma_0)\mathbb{E}(\tilde{R}_2) = D$$

[1] If $\epsilon < \frac{(\gamma_1-\gamma_0)\mathbb{E}(\tilde{R}_2)-D}{\frac{L(1-p)}{p}+\mathbb{E}(\tilde{R}_2)}$, then $(\gamma_1-\gamma_0)\mathbb{E}(\tilde{R}_2) - \epsilon\mathbb{E}(\tilde{R}_2) > D + \frac{\epsilon L(1-p)}{p}$. Therefore, $(\gamma_1-\gamma_0-\epsilon)\mathbb{E}(\tilde{R}_2) = (\gamma_1-\gamma_0)\mathbb{E}(\tilde{R}_2) - \epsilon\mathbb{E}(\tilde{R}_2) > D + \frac{\epsilon L(1-p)}{p}$.

Threat of termination

- The probability of liquidation is

$$1 - \gamma_0 = \frac{I - A}{p\mathbb{E}(\tilde{R}_2) + (1-p)L}$$

- Liquidation (which is inefficient) is less likely when
  - The gain of continuing, $\mathbb{E}(\tilde{R}_2)$, is large: the borrower has more to lose by not honoring the payment D
  - The liquidation value L increases: it alleviates the break-even condition of the lender
  - The probability p of the first-period success increases: lenders are repaid more often
  - The borrower net worth increases: the amount borrowed, $I - A$, decreases

Threat of termination

- Lenders have to incentivize repayment by engaging in some wasteful (inefficient) activity
  - In this model, the inefficiency comes from the inefficient liquidation, as $\mathbb{E}(\tilde{R}_2) > L$
- Costly collateral pledging interpretation: we can interpret L as the value that the lender obtains by seizing some collateral (an asset). Inefficiency comes from the fact that this asset is worth more when it is in the borrower's hands: $\mathbb{E}(\tilde{R}_2) > L$
- Room for renegotiation: after the termination decision, but before its implementation, the borrower can offer a bribe slight above L (and smaller than $\mathbb{E}(\tilde{R}_2)$ to the lender, who would be happy to accept. This prospect ex ante undermines incentives to pay out income, and reduces welfare

Sovereign debt

- Eaton and Gersovitz (1981), Arellano (2008)
- Small open economy has endowments that generate a nonstorable and stochastic output $\tilde{y}_t \in \mathbb{Y} = [\underline{y}, \bar{y}]$
- The stochastic process $(\tilde{y}_t)_{t=0,1,2,..}$ is a Markov process with a transition function $f(y', y)$ giving the density of next period's output y' given that current period's output is y
- The country has preferences

$$\mathbb{E}_0\left(\sum_{t=0}^{+\infty}\beta^t u(c_t)\right)$$

where $\beta \in (0,1)$, $u' > 0$ and $u'' < 0$

- The country can smooth consumption by borrowing to or lending from foreign investors
- External capital markets are competitive; investors are risk-neutral
- Debt contracts (bonds) have a maturity of one period and are not state-contingent (i.e. markets are incomplete)
- Debt contracts are not enforceable, that is, at any time the country can renege on its debt (even if the country has the means to repay)
- Default has costs: output loss and exclusion from capital markets
- The country defaults whenever this option is more valuable than repaying

- In equilibrium, endogenous prices (equivalently, interest rates) will reflect the probability of default

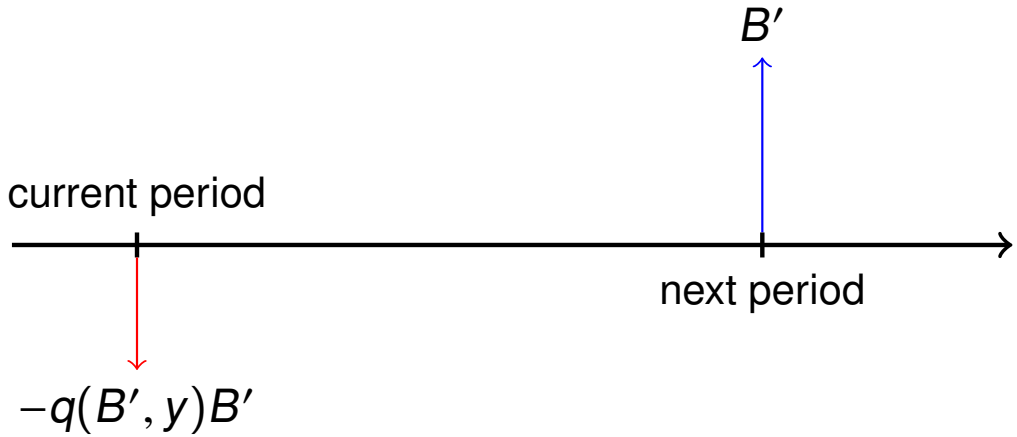
- We denote the *unit* price a bond with face value B' by $q(B', y)$, a function of the current output y and the bond size B' (determinants of government's incentives to default)

- The price of a bond with face value B' is then $q(B', y)B'$

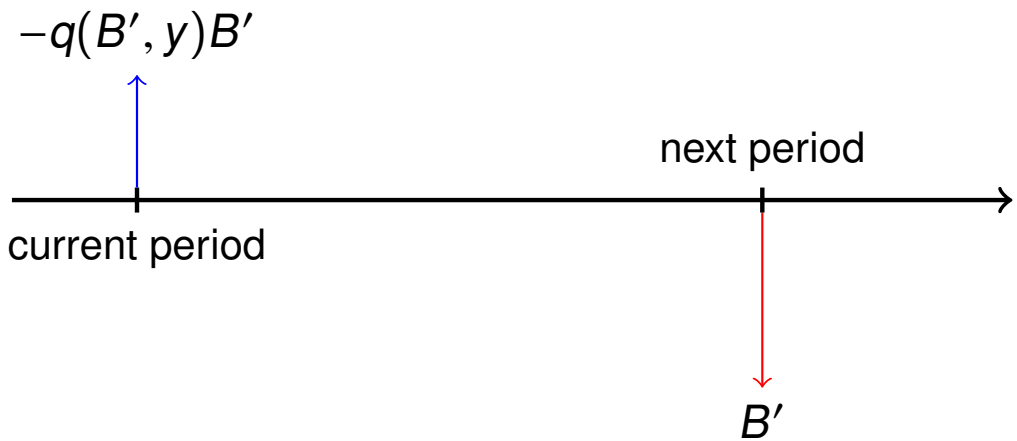
- Alternatively, the interest rate on a bond with face value B' is

$$1 + r(B', y) = \frac{B'}{q(B', y)B'} = \frac{1}{q(B', y)}$$

- When the country buys bonds with face value $B' > 0$ (i.e. lends), it pays $q(B', y)B'$ in the current period and receives B' one period later



- When the country buys bonds with face value $B' < 0$ (i.e. sells bonds with face value $-B' > 0$ or borrows), it receives $-q(B', y)B'$ in the current period and pays $-B'$ one period later



- The country starts with initial assets (debt) B and observes the endowment realization y

- When the country chooses to repay its debts, the resource constraint is

$$c = y + B - q(B', y)B'$$

- When the country defaults, it loses access to foreign credit markets (autarky) for a stochastic number of periods

- The country will regain access to international markets next period with a probability θ

- Default also causes a reduction in output during the periods the government is in autarky

- Instead of consuming the endowment y, the country consumes $y^{def} = h(y) \leq y$ until it reenters financial markets

- Foreign investors can invest their funds at a constant international interest rate $r > 0$

- Investors can observe y every period and know the stochastic process

- Competition and risk-neutrality imply that, given a default probability $\delta(B', y)$, which itself is a function of y and B'

$$q(B', y)B'(1 + r) = B'[1 - \delta(B', y)] \quad \Rightarrow \quad q(B', y) = \frac{1 - \delta(B', y)}{1 + r}$$

- When $B' > 0$ (the country is lending), the default probability is zero, but when $B' < 0$ (the country is borrowing) $\delta(B', y) \in [0, 1]$

- Prices decrease (equivalently, interest rates increase) then the default probability increases

Recursive equilibrium

- Given the state variables $s = (B, y)$, the policy functions $B'(s)$ for government assets, $q(s) = q(B'(s), y)$ for bond prices and $c(s)$ for consumption determine the equilibrium
- Define $V^o(B, y)$ as the value function for the country that has the option to default
- Let $V^d(B, y)$ denote the value function when the country chooses to default, and let $V^c(B, y)$ denote the value function when the country chooses to repay
- Then

$$V^o(B, y) = \max \left\{ V^d(B, y), V^c(B, y) \right\}$$

- Value of default is

$$V^d(y) = u(y^{def}) + \beta \int_{y'} \Big[\theta V^o(0, y') + (1-\theta) V^d(y')\Big] f(y', y) dy'$$

- The value of repayment is

$$V^c(B, y) = \max_{B' \geq Z} \left\{ u\Big(y + B - q(B', y)B'\Big) + \beta \int_{y'} V^o(B', y') f(y', y) dy' \right\}$$

where $Z < 0$ is a lower bound on debt that prevents Ponzi schemes

- Default is optimal if $V^d(y) > V^c(B, y)$

- At any given period, the set of $y \in \mathbb{Y}$ for which default takes place depends on the level of government's assets maturing at the period B, and is given by

$$D(B) = \left\{ y \in \mathbb{Y} : V^d(y) > V^c(B, y) \right\}$$

- The probability that country defaults next period when it contracts debt B' is

$$\delta(B', y) = Pr[y' \in D(B')|y] = \int_{D(B')} f(y', y)dy'$$

Definition

The recursive equilibrium for this economy is defined by a set of policy functions, defined over the state variables $s = (B, y)$, for (i) consumption $c(s)$; (ii) government's assets holdings $B'(s)$ and default sets $D(B)$; and price functions for bonds $q(B', y)$ such that:

1. Consumption $c(s)$ satisfies the resource constraint

2. Taking as given the bond price function $q(B', y)$, asset's holdings $B'(s)$ and the default set $D(B)$ maximize the country's utility

3. Bond prices $q(B', y)$ reflect the government's default probabilities and are consistent with zero profits

- **Claim 1:** For a given $y \in \mathbb{Y}$, $V^c(y, B)$ is increasing in B

  Suppose that $B_2 \geq B_1$. Let B_1' and B_2' be the optimal choices when $B = B_1$ and $B = B_2$, respectively. Then

$$\begin{aligned} V^c(y, B_2) &= u\Big(y + B_2 - q(B_2', y)B_2'\Big) + \beta \mathbb{E}_y\Big[V^o(y', B_2')\Big] \\ &\geq u\Big(y + B_2 - q(B_1', y)B_1'\Big) + \beta \mathbb{E}_y\Big[V^o(y', B_1')\Big] \\ &\geq u\Big(y + B_1 - q(B_1', y)B_1'\Big) + \beta \mathbb{E}_y\Big[V^o(y', B_1')\Big] \\ &= V^c(y, B_1) \end{aligned}$$

- **Claim 2:** If $B_2 \geq B_1$, then $D(B_2) \subseteq D(B_1)$

  As $V^c(y, B)$ increases with B, while $V^d(y)$ remains constant, default is more likely when B is smaller (the level of debt is higher)

- As

$$\delta(B', y) = Pr[y' \in D(B')|y] = \int_{D(B')} f(y', y)dy'$$

and

$$q(B', y) = \frac{1 - \delta(B', y)}{1 + r}$$

by Claim 2 it follows that the equilibrium price function $q(B', y)$ is increasing in B' (and the interest rate is decreasing)

- Let $\underline{B}$ be the upper bound of assets for which the default set is the entire set $\mathbb{Y}$, and let $\bar{B}$ be the lower bound of assets for which the default set is empty, where $\underline{B} \leq \bar{B} \leq 0$ (due to claim 1):

$$\underline{B} = \sup \{B : D(B) = \mathbb{Y}\}$$
$$\bar{B} = \inf \{B : D(B) = \emptyset\}$$

The i.i.d case when $\theta = 0$ and $h(y) = y$

- **Claim 3:** If, for some B, the default set is non-empty, $D(B) \neq \emptyset$, then there are no contracts available $\{q(B'), B'\}$ such that the economy can experience capital inflows, $B - q(B')B' > 0$

  Suppose by contradiction that $\exists \hat{B}$ such that $B - q(\hat{B})\hat{B} > 0$. Then, for a y for which default is optimal

$$
\begin{aligned}
& V^d(y) > V^c(y, B) \\
& \iff u(y) + \beta \mathbb{E}\Big[V^d(y')\Big] > u\Big(y + B - q(\hat{B})\hat{B}\Big) + \beta \mathbb{E}\Big[V^o(\hat{B}, y')\Big] \\
& \iff u(y) - u(y + B - q(\hat{B})\hat{B}) > \beta \mathbb{E}\Big[V^o(\hat{B}, y')\Big] - \beta \mathbb{E}\Big[V^d(y')\Big] \geq 0
\end{aligned}
$$

  which is a contradiction as $u(.)$ is increasing

- If the country could roll over the debt, that is, borrow an amount large enough to repay the previous debt, it would choose to do so, given that it can still exercise the option to default in the next period

- **Claim 4:** Default incentives are stronger the lower the endowment. For all $y_1 \leq y_2$, if $y_2 \in D(B)$ then $y_1 \in D(B)$

  Let B_1' and B_2' be the optimal asset choices in case of repayment when $y = y_1$ and $y = y_2$, respectively. If $y_2 \in D(B)$, then

$$u(y_2) + \beta\mathbb{E}\Big[V^d(y')\Big] > u\Big(y_2 + B - q(B_1')B_1'\Big) + \beta\mathbb{E}\Big[V^o(B_1', y')\Big]$$

  or

$$u(y_2) - u\Big(y_2 + B - q(B_1')B_1'\Big) > \beta\mathbb{E}\Big[V^o(B_1', y')\Big] - \beta\mathbb{E}\Big[V^d(y')\Big] \quad (1)$$

  Since $u(.)$ is concave, $y_1 \leq y_2$, and, from Claim 3, $B - q(B_1')B_1' < 0$,

$$u(y_1) - u\Big(y_1 + B - q(B_1')B_1'\Big) \geq u(y_2) - u\Big(y_2 + B - q(B_1')B_1'\Big) \quad (2)$$

  Combining (1) and (2)

$$u(y_1) - u\Big(y_1 + B - q(B_1')B_1'\Big) > \beta\mathbb{E}\Big[V^o(B_1', y')\Big] - \beta\mathbb{E}\Big[V^d(y')\Big]$$
$$\Rightarrow u(y_1) + \beta\mathbb{E}\Big[V^d(y')\Big] > u\Big(y_1 + B - q(B_1')B_1'\Big) + \beta\mathbb{E}\Big[V^o(B_1', y')\Big]$$

- Net repayment is more costly when income is low because $u(.)$ is concave [$u(x) - u(x-k)$ increases when x decreases]

- The claims imply that default happens when debt levels are high, the country is in a recession and cannot roll over its debt

- **Corollary**: for $\underline{B} < B < \bar{B}$, there exists a threshold $y^*(B)$ such that default set is
$$D(B) = \left[\underline{y}, y^*(B)\right)$$

- Given a choice B', the default probability is $F\left(y^*(B')\right)$ where F is the cdf of $\tilde{y}_t$ and thus the price function is
$$q(B') = \frac{1 - F\left(y^*(B')\right)}{1+r}$$
which does not depend on current endowment y given the i.i.d assumption

- Note, however, that equilibrium prices depend on y as the optimal choice of B' depends on y

- The maximum increase in consumption in a given period is given by a level B^* that solves

$$\min_{B'} q(B')B' = \frac{1 - F\Big(y^*(B')\Big)}{1+r}B'$$

  Such level exits as bond prices go to zero as debt increases

- This endogenous borrowing limit implies that the country faces a limited set of feasible consumption levels

- Given a bad shock, a country would like to borrow more but such a contract is not available

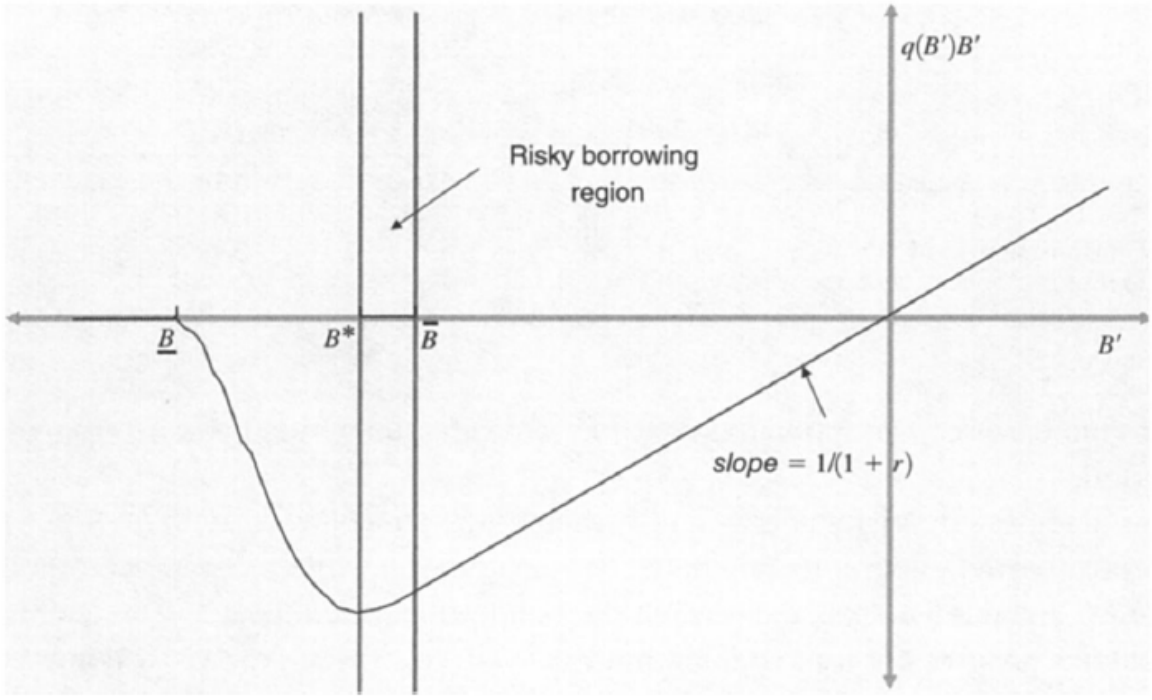


FIGURE 2. TOTAL RESOURCES BORROWED

- It is never optimal to choose $B' < B^*$ (borrow more than $-B^*$) as the country can raise the same amount with a higher price (and thus a smaller repayment next period)
- When $B > \bar{B}$, debt is risk-free and $q(B) = 1/(1+r)$

References I

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